Air Quality Prediction

CSAI Project 19 — Phase 1 Machine Learning

Project 19 Team

What we want to predict

- Predict PM2.5 one hour ahead, in $\mu g/m^3$.
- After the 10:00 reading, predict the concentration at 11:00.
- Use pollution and weather through 10:00, plus the known calendar.
- Compare XGBoost with persistence, daily seasonal naive, and Ridge.

Our question: how much does this added complexity improve on the latest reading?

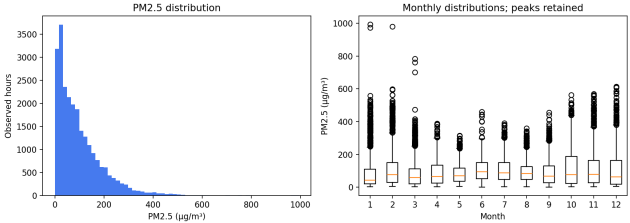
The supplied data and our cleaning decisions

- UCI Beijing PM2.5: 43,824 hourly rows, January 2010–December 2014.
- Sorted, unique timestamps; no missing timestamp rows.
- 2,067 missing PM2.5 readings (4.72%); weather fields are complete.
- 214 PM2.5 gaps; the longest lasts 155 hours.
- Keep missing measurements and valid peaks. Exclude the row ID.
- Pollution and airport weather come from different locations.

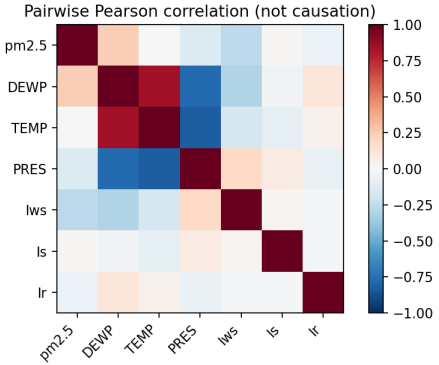
Missing targets are never filled for scoring. Ridge imputes inputs inside each training fold; XGBoost uses missing-value splits.

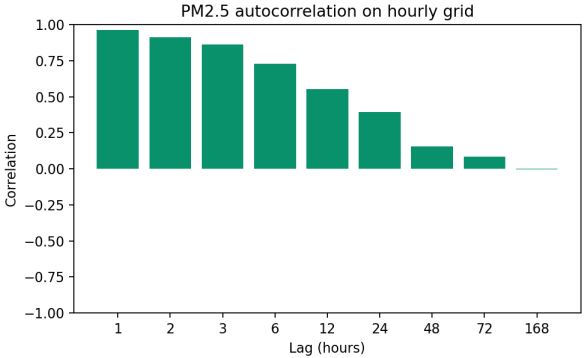
Training EDA: the upper tail matters

- Only 2010–2012: 24,418 measured concentrations.
- Missing PM2.5: 7.17%.
- Mean 97.82; median 73.00.
- Range 0–994; training p90 = 218.
- Retain 797 readings above the IQR fence of 305.50.



What recent history and weather tell us





- PM2.5 autocorrelation: 0.961 at 1 hour; 0.394 at 24 hours; about −0.006 at 168 hours.
- Accumulated wind-speed correlation: −0.264. Association does not establish causation.

Turning observations into forecast inputs

- PM2.5 lags: 0, 1, 2, 3, 6, 12, 24, 48, 72 and 168 hours.
- Trailing means, standard deviations and observed fractions.
- Known target hour, weekday and month: sine/cosine; weekend flag.
- Numeric weather at offsets 0, 1 and 24; current wind direction.
- Missingness stays visible. All windows end at origin t.

70 columns before wind encoding; 73 after encoding the four observed categories.
Lag zero is valid: our label is $y(t+1)$, while the latest input is $y(t)$.

Why these models and references?

- Persistence: $\hat{y}_{t+1|t} = y_t$.
- Daily seasonal naive: $\hat{y}_{t+1|t} = y_{t-23}$.
- Ridge: squared error with an L2 penalty, fixed $\alpha = 10$.
- XGBoost: additive trees with gradient/Hessian updates and complexity penalties.
- The readings connect to Liang et al. (pollution/weather), Chen and Guestrin (boosting), Hyndman and Athanasopoulos (evaluation), and Hoerl and Kennard (Ridge).

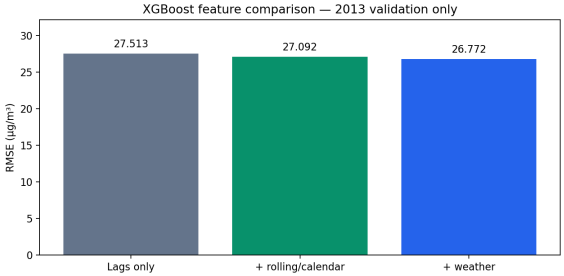
How we keep the comparison chronological

2010–2012	Initial training and EDA
2013	Four expanding validation quarters
2014	Four backtest quarters; frozen settings

- Fit once per quarter using labels available by the first origin.
- Keep the fit fixed within each quarter; update observed histories hourly.
- Later quarters can use earlier observed quarters.
- Every method scores the same observed targets and baseline inputs.
- 2014 coverage: 8,531 / 8,760 hours, or 97.39%.

Parameter selection and feature comparison

- Three candidates: validation RMSE 26.772, 26.801 and 26.953.
- Selected: 300 trees, depth 3, rate 0.05, minimum child weight 5.
- Row/column sampling 0.8/0.9, L2 = 5, seed 42.
- Feature groups use these same settings and 2013 dates only.



Full-feature XGBoost still trails validation persistence (24.870 RMSE) and Ridge (23.809).

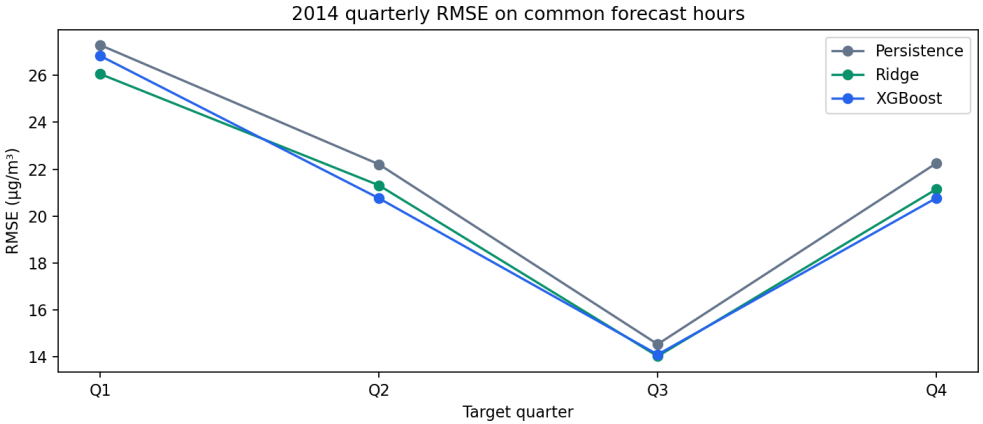
Measured results on 2014

Model	MAE	RMSE	R^2
persistence	11.926	22.033	0.944
ridge	11.825	21.063	0.949
seasonal naive	67.296	98.754	-0.121
xgboost	11.833	21.088	0.949

MAE and RMSE are in $\mu g/m^3$; all rows use 8,531 common hours.

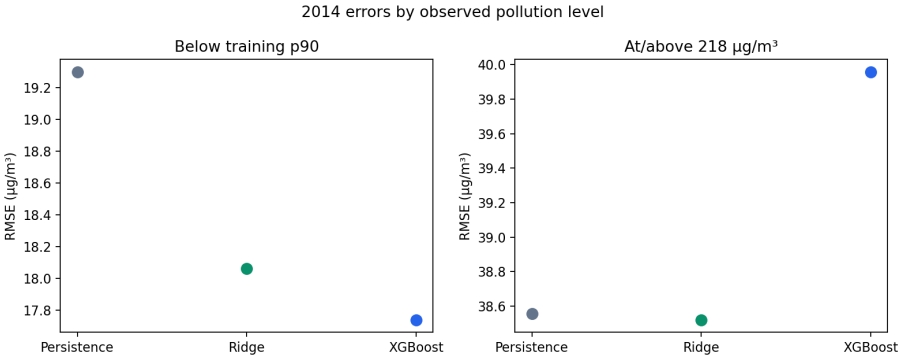
- XGBoost improves RMSE over persistence by 4.29%; Ridge by 4.41%.
- Ridge is slightly better overall: its RMSE is lower by only 0.025.
- No statistical significance claim is made for that small difference.

The ranking changes across quarters



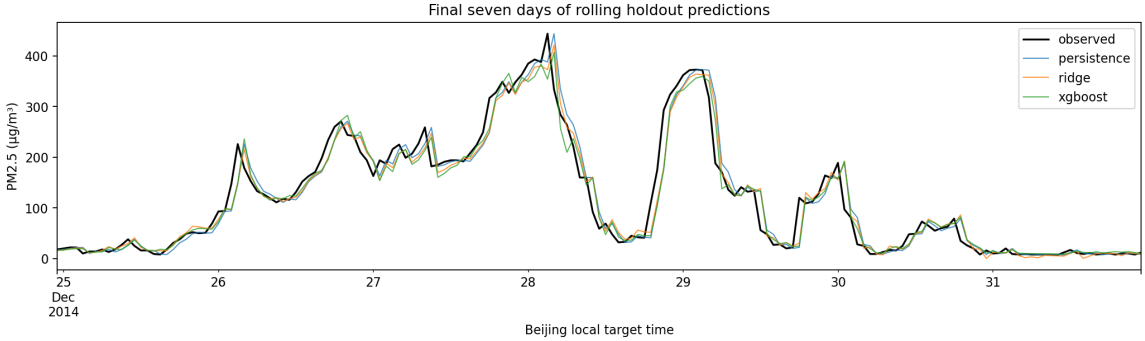
Ridge is better in Q1/Q3; XGBoost in Q2/Q4. Both improve RMSE over persistence in each quarter.

High-pollution hours remain difficult



- At/above training p90 = 218: 865 common hours.
- XGBoost RMSE 39.959 versus persistence 38.557; peak bias is −11.912.
- Its overall improvement does not establish better peak forecasting.

A week of historical backtest forecasts



The saved deployment model is a later refit. It was not used to produce these backtest predictions.

What we can reproduce, and what remains uncertain

- Saved input/source hashes, versions, configuration, selection record and forecast rows.
- Cleaned hourly export, gap dates, feature dictionary and worked feature preview.
- Residual lag-1 correlation: XGBoost 0.120; Ridge 0.063.
- Historical location only; weather publication delays are assumed.
- No causal weather claim, calibrated uncertainty or daily AQI label.
- Holdout results have been viewed; further tuning needs fresh evaluation.

What we learned

- The latest reading is a strong one-hour forecast.
- Our XGBoost and Ridge models improve pooled RMSE modestly in 2014.
- Temporal and weather features help this XGBoost setup on validation dates.
- Ridge remains a competitive, slightly stronger overall comparator.
- Peak prediction needs further work; the pooled score hides that weakness.

This completes the classical ML experiment for Phase 1. Deep learning and a hybrid model belong to the later phase.